GigaWhat Energy

GigaWhat Gas
Controlled procedure

PR-GAS-071 Investigating Gas in Buildings

WITHDRAWN: DO NOT USE. CHECK THE DOCUMENT REGISTER.

Field	Value
Document ID	PR-GAS-071
Version	2
Effective from	2021-05-04
Review due	2024-05-04
Owner	Head of Gas Network Operations
Approved by	Gas Safety Manager
Applies to	All GigaWhat Gas sites: low pressure mains

FICTIONAL DOCUMENT. Created for the GigaWhat demo. Not for real operational use.

1 Purpose

This procedure sets out how GigaWhat Gas investigates gas found inside a building that is entering from an external source, such as a leaking low pressure main or service. It covers confirming that gas is entering from outside, evacuation, checking the building and its neighbours, tracing the source, and controlling the ingress until the repair is complete.

2 Scope

This procedure applies when a First Call Operative attending a reported escape under PR-GAS-003 finds gas in a building with the emergency control valve (ECV) turned off, or finds gas in a building with no gas supply.

It applies to buildings near low pressure (LP) mains and services, including MAIN-LP-17 (Mill Lane, PE) and MAIN-LP-22 (Station Road, cast iron). Cast iron mains such as MAIN-LP-22 are prone to joint leakage and fractures, particularly after frost and ground movement, and gas from them can travel long distances through the ground, ducts and drains.

3 References

- PR-GAS-003 Response to Reported Gas Escapes
- PR-GAS-021 Permit-to-Work for Gas Operations
- PR-GAS-040 Working in Hazardous Areas (DSEAR Zones)

- PR-GAS-052 Safe Excavation Near Live Gas Mains
- PR-CORP-001 Reporting Incidents and Near Misses
- PR-CORP-004 Lone Working
- Gas Safety (Management) Regulations 1996 and HSE guidance L80
- Dangerous Substances and Explosive Atmospheres Regulations 2002 (DSEAR) and HSE guidance L138

4 Definitions

Gas in buildings: Gas found inside a building that is entering from an external source rather than from the customer's installation.

Ingress point: A place where gas enters a building, such as a service entry, duct, drain, cellar wall or floor void.

LEL: Lower explosive limit, about 5% gas-in-air for natural gas. 20% LEL equals 1% gas-in-air.

Bar hole: A narrow hole driven into the ground with a bar to sample gas below the surface.

Readings plot: A sketch of the building and surrounding ground showing each reading, its location and time.

5 Responsibilities

First Call Operative: Confirms gas in buildings, decides on immediate evacuation, carries out the checks in this procedure and keeps the Gas Network Controller informed.

Authorised Person (Gas): Attends every gas-in-buildings incident, takes charge on arrival, decides on further evacuation and authorises re-occupation.

Gas Network Controller: Sends the Authorised Person (Gas) and a repair team, records evacuations and informs the Gas Safety Manager.

Gas Safety Manager: Is informed of every evacuation and decides on any wider response, including leakage survey of nearby mains.

6 Procedure

6.1 Confirming gas in buildings

6.1.1 Confirm that the ECV is off and that the customer's installation has been made safe under PR-GAS-003.

6.1.2 Ventilate the building by opening doors and windows, then re-test after 10 minutes.

6.1.3 Treat the incident as gas in buildings if gas is still detected with the ECV off, or if readings are highest at an ingress point rather than near appliances or pipework.

6.1.4 Tell the Gas Network Controller that you have gas in a building from a suspected external source, and ask for the on-call Authorised Person (Gas) to attend.

6.1.5 Give the Gas Network Controller the nearest main if known, for example MAIN-LP-22.

6.2 Evacuation

> WARNING: Gas entering from the ground can build up quickly in cellars and voids and reach the flammable range without warning. Do not delay evacuation to complete the investigation.

6.2.1 Evacuate the building immediately if any reading inside is 20% LEL or above, or if readings are rising.

6.2.2 Evacuate any building where gas is found in a cellar, void or room at 20% LEL or above.

6.2.3 Evacuate adjoining properties that share walls, cellars or ducts with an affected building.

6.2.4 Send evacuated people to a safe place upwind and at least 50 m away, and check that everyone is accounted for.

6.2.5 Tell the Gas Network Controller of every evacuation, and ask for police and fire service support through 999 if more than one property is evacuated.

6.2.6 Where readings are below 20% LEL and falling, keep occupants out of affected rooms and re-test every 15 minutes.

6.3 Checking the building and its neighbours

> WARNING: Do not operate electrical switches, doorbells or non-intrinsically safe equipment in an affected building. Use only equipment permitted by PR-GAS-040.

6.3.1 Test every room, cupboard and cellar, concentrating on points where gas, water, electricity, telecoms and drainage services enter.

6.3.2 Test under-floor voids through airbricks or lifted floorboards, and wall cavities through airbricks or weep holes.

6.3.3 Test the adjoining properties on both sides and the properties opposite, including their cellars and voids.

6.3.4 Ventilate affected voids and cellars by opening airbricks, doors and hatches where safe to do so.

6.3.5 Record each reading, its location and time on the readings plot.

6.3.6 Re-test all positive locations every 15 minutes until the Authorised Person (Gas) takes charge.

6.4 Tracing the external source

> CAUTION: Bar holes can strike cables and pipes. Scan with a cable avoidance tool and follow PR-GAS-052 before driving any bar hole.

6.4.1 Test drains, inspection chambers, utility boxes, valve covers and cracks in the road or footway near the building.

6.4.2 Drive bar holes about 450 mm deep at no more than 2 m intervals along the line of the service and the main, keeping at least 150 mm above the known depth of any pipe.

6.4.3 Test each bar hole and add the readings to the readings plot.

6.4.4 Identify the highest concentration on the plot as the likely source, and extend the bar holes along the main for 10 m either side of it.

6.4.5 Where the main is cast iron, check for gas along the whole length between the building and the nearest joints on either side.

6.4.6 Give the readings plot to the Authorised Person (Gas) and the repair team on arrival.

6.5 Controlling the ingress

6.5.1 Leave bar holes open between the source and the building so gas vents to atmosphere rather than into the building.

6.5.2 Where readings at the building are not falling, drive extra bar holes along the building wall to intercept the gas.

6.5.3 Keep drain and chamber covers open, guarded and attended where they are venting gas.

6.5.4 Keep checking the building and its neighbours every 15 minutes until the repair is complete.

> HOLD POINT: The repair team must not excavate at the source until the Authorised Person (Gas) has issued a permit-to-work under PR-GAS-021 and confirmed the excavation controls in PR-GAS-052.

6.6 Repair and re-occupation

6.6.1 The Authorised Person (Gas) takes charge of the site on arrival and receives a briefing from the First Call Operative.

6.6.2 After the repair, re-test every bar hole, drain and chamber used during the investigation.

6.6.3 Re-test every room, void and cellar in each affected building.

6.6.4 Continue ventilating until no gas has been detected anywhere in the building for at least 30 minutes.

> HOLD POINT: Do not allow anyone back into an evacuated building until the Authorised Person (Gas) has confirmed that the source is repaired and no gas has been detected in any room, void or cellar for at least 30 minutes.

6.6.5 Restore gas supplies only after re-occupation has been authorised, testing each installation for tightness before relighting appliances.

6.7 Emergency

6.7.1 Call 999 for fire, explosion or injury.

6.7.2 If you cannot contact the Gas Network Controller by radio, call the National Gas Emergency Service on 0800 111 999 and ask to be connected to the GigaWhat Gas Network Controller.

6.7.3 Keep lone working check-ins under PR-CORP-004 until another operative or the Authorised Person (Gas) arrives.

6.7.4 Report every evacuation, fire, explosion and injury under PR-CORP-001.

7 Records

The readings plot, evacuation details and re-occupation time are attached to the Gas Network Controller's job record. The Authorised Person (Gas) records the re-occupation authorisation and the repair permit number on the job record. Records are kept for 10 years.

8 Revision history

Version	Date	Change
1	2017-09-04	First issue. Gas-in-buildings investigation separated from PR-GAS-003.
2	2021-05-04	Added checks of properties opposite, cast iron joint checks along MAIN-LP-22 type mains, and hold point before excavation at the source.